

Python 3.12 Environment Setup Guide for AI & Agentic AI Labs

Objective

This guide helps students install Python 3.12, create a virtual environment, and configure Visual Studio Code for all hands-on sessions involving:

- Python Programming
 - Generative AI
 - LangChain
 - LangGraph
 - Ollama
 - Agentic AI Frameworks
 - Machine Learning Labs
-

Prerequisites

Before starting, ensure you have:

1. Windows 10 or Windows 11
2. Administrator access to your machine
3. Internet connection
4. Visual Studio Code installed

Recommended VS Code Extensions:

- Python
 - Pylance
 - Jupyter
-

Step 1: Install Python 3.12

Download Python 3.12

Visit:

<https://www.python.org/downloads/>

Download the latest Python 3.12.x version.

During Installation

- ✓ Add Python to PATH
 - ✓ Install launcher for all users
 - ✓ Customize Installation (Optional)
 - ✓ Install for all users
-

Step 2: Verify Python Installation

Open PowerShell and run:

```
python --version
```

or

```
py --version
```

Expected Output:

```
Python 3.12.x
```

If multiple versions exist:

```
py -0
```

Example:

```
Installed Pythons found by py Launcher
```

```
-V:3.14
```

```
-V:3.12
```

Step 3: Create Project Folder

Open PowerShell and execute:

```
mkdir frameworks  
cd frameworks
```

Step 4: Create Virtual Environment

Create a Python 3.12 virtual environment:

```
py -3.12 -m venv venv
```

This creates the following structure:

```
frameworks/  
|  
├─ venv/  
|   ├─ Scripts/  
|   ├─ Include/  
|   ├─ Lib/  
|   └─ pyenvn.cfg
```

Step 5: Activate Virtual Environment

PowerShell

```
.\venv\Scripts\Activate.ps1
```

Expected Output:

```
(venv) PS C:\Users\Student\frameworks>
```

Step 6: Verify Python Version

After activation:

```
python --version
```

Expected Output:

```
Python 3.12.x
```

To verify environment location:

```
where python
```

Expected Output:

```
C:\Users\Student\frameworks\venv\Scripts\python.exe
```

Step 7: Open Project in VS Code

From PowerShell:

```
code .
```

This opens the current project folder in Visual Studio Code.

Step 8: Select Python Interpreter

Inside VS Code:

1. Press Ctrl + Shift + P
2. Search for:

Python: Select Interpreter

1. Choose:

venv\Scripts\python.exe

or

Python 3.12 (venv)

Step 9: Install Required Packages

Upgrade pip:

```
python -m pip install --upgrade pip
```

Verify:

```
pip --version
```

Step 10: Install AI Development Libraries

Install commonly used packages:

```
pip install langchain
pip install langgraph
pip install ollama
pip install openai
pip install requests
pip install pandas
pip install numpy
pip install jupyter
```

Install all at once:

```
pip install langchain langgraph ollama openai requests pandas numpy jupyter
```

Verify Installation

Check installed packages:

```
pip list
```

Check LangChain:

```
pip show langchain
```

Check LangGraph:

```
pip show langgraph
```

Test Your Environment

Create a file named:

```
test.py
```

Add the following code:

```
print("Python Environment Ready!")
```

Run:

```
python test.py
```

Expected Output:

Python Environment Ready!

Common Commands

Activate Environment

```
.\venv\Scripts\Activate.ps1
```

Deactivate Environment

```
deactivate
```

Install Package

```
pip install package_name
```

Upgrade Package

```
pip install --upgrade package_name
```

View Installed Packages

```
pip list
```

Troubleshooting

Issue 1: Activation Disabled

Error:

```
running scripts is disabled on this system
```

Solution:

Open PowerShell as Administrator:

```
Set-ExecutionPolicy RemoteSigned
```

Choose:

```
Y
```

Restart PowerShell.

Issue 2: Python Not Found

Error:

```
python is not recognized
```

Solution:

Reinstall Python and ensure:

✓ Add Python to PATH is selected

Issue 3: Wrong Python Version

Verify:

```
python --version
```

If incorrect:

```
py -3.12 -m venv venv
```

Recreate the virtual environment.

Expected Setup at the End

```
frameworks/  
|  
├─ venv/  
├─ test.py  
└─ requirements.txt
```

PowerShell Prompt:

```
(venv) PS C:\Users\Student\frameworks>
```

Python Version:

```
Python 3.12.x
```

Status:

- ✓ Python Installed
- ✓ Virtual Environment Created
- ✓ VS Code Configured
- ✓ AI Libraries Installed
- ✓ Ready for LangChain, LangGraph and Agentic AI Labs